TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Safety-based Human-Robot-Collaboration Framework for Vision-Language-Action Controlled Humanoid Robots			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	Human Resource Competency Development Services	Human–System Interaction Technologies
Concept of Proposed Technology	The proposed technology is a two-layer framework that combines VLA-based, risk-aware task planning with RL-based motion control and a deterministic safety shield. The aim is to close the current gap between semantic planning (text/image) and safe physical execution on humanoid robots, explicitly demonstrating ISO/TS 15066-compliant safe touch properties in shared workspaces rather than just providing prototypical demos. **Core concept:** Hierarchical two-level architecture • *Higher layer (low frequency ~ 5 Hz)*: VLA task manager with task planner, target generator and safety manager (performs risk-aware task scheduling and generates low-collision target states by dynamically replanning task sequences when the lower layer exceeds thresholds regarding to ISO/TS 15066). • *Lower layer (high frequency ~ 250 Hz):* RL motion policy pool and Sim2Real pipeline under			

	a high-frequency safety shield (enforces safe trajectory execution through deterministic monitoring of predicted collision severity, real-time speed scaling, and emergency braking).
	Safety architecture: Hybrid approach ensuring verifiable safety guarantees independent of ML uncertainties
	• *Semantic layer*: CAD-based digital twin risk assessment (planning validation)
	• *Control layer*: Non-AI-based environment monitoring via safety-rated LIDAR zones for real-time human presence detection, e.g. using SafeVisionary2 with PLc
	Main functions:
	• **Semantic layer:**
	  ○ *Task Planner*: Processing of natural language and visual information for situation-aware task decomposition for humanoids.
	  ○ *Target Generator*: Generation of end-effector or joint-based target poses from abstract commands, considering the current environment.
	• **Control layer**: Execution of movements via RL policies with online monitoring by the Safety Shield with automatic speed/trajectory adjustment.
	• **Cross layer**: Consistent risk assessment according to ISO/TS 15066 from the planning to the execution and monitoring levels.
	Potential areas of application:
	• The architecture is designed for cross-domain applicability and can be tailored to partner-specific use cases – we actively seek application partners to co-validate additional scenarios.
	• *Initial validation focus:* Human-humanoid cooperation in service and assembly scenarios where safety-critical physical contact must not only be avoided but also controlled in extreme cases, e. g.:
	  ○ Collaborative joining and insertion activities in manufacturing in shared workspaces.
	  ○ Flexible logistics and handling tasks in unstructured environments (variety of variants, changing layouts).

Rationale for Joint R&D Project	The project addresses a problem that no single actor can solve alone: the state of the art in VLA-based robotics systems provides neither explicit, ISO-compliant safety guarantees for HRC nor robust Sim2Real-capable motion strategies for humanoids in unstructured environments. A joint R&D project is needed to systematically close the gap between semantic VLA planning and safety-critical physical execution, while paving the way for internationally compatible safety standards and industrially usable “safety-based VLA” middleware. **Technical necessity:** Current VLA approaches do not meet key HRC requirements (no verifiable ISO/TS 15066 compliance, insufficient force/contact monitoring, weak coupling between speech/image planning and precise manipulation) [1]. These gaps can only be planned, modelled and experimentally validated in integrated architecture. **Complementary expertise:** Required are proven safety intelligence for collaborative robots (simulation-based risk assessment, safety shield, Sim2Real) and an applied research institute with experience in VLM/VLA and task planning in production. Only this combination enables an HRC+

	framework that is both cognitively powerful and physically secure. **Technological sovereignty:** This German-Korea partnership is essential not only for technical reasons but also to establish technological sovereignty in safety-critical humanoid systems. Developing our own safety middleware instead of relying on black-box solutions from USA/China, the partnership ensures that both Germany and Korean industries can deploy humanoid systems without dependency on external tech providers. Both regions prioritize reducing critical technology dependencies in AI and advanced robotics (Germany's High-Tech Agenda [2], Korea's National Strategic Technology Policy [3]) – this project directly contributes to these goals by creating an certifiable and industrially deployable alternative to current market-dominant systems. **Standardization and market access:** The coupling of ISO-related safety research (e.g. participation in robot safety committees) with real requirements from European and Korean industrial applications is a prerequisite for directly transferring results into future humanoid standards and a certifiable "safety-based VLA SDK". The proposed architecture ensures compliance with the EU AI Act [4] by treating the control layer as a high-risk safety component under the Machinery Directive [5], implementing the biomechanical constraints defined in ISO/TS 15066 [6] directly into the VLA execution loop. **Risk reduction and validation:** Distributed test beds (cobot cell, humanoid platform, industrial joining scenarios) in several countries enable more robust evidence on safety and performance limits than an isolated laboratory project and reduce the technical and regulatory risk for later commercialization.
Key Words	1. Humanoid robots 2. Vision-Language-Action 3. Human-Robot-Collaboration 4. Robot Safety 5. Reinforcement Learning

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (5)
R&D Objectives	The project aims to realize one of the first industrial-grade pilot installations for safe human-humanoid collaboration in manufacturing, demonstrating ISO/TS 15066-compliant physical interaction in a real assembly scenario. This pilot serves as a validated reference system for future commercial deployments and as the technical foundation for a certifiable "safety-based VLA SDK" for humanoid systems. To achieve this application-driven objective, the development and integration of the following technical subsystems are subgoals for the pilot installation: • Development of a risk-aware VLA task manager (planner, dynamic replanning) that translates speech and image information into collaborative, safety-assessed work steps for robots. • Development of a vision-language-action target generator that converts temporal plans and environmental images into low-collision target states (pose/target configuration) for robot systems. • Implementation of a real-time safety shield with collision severity/probability predictors, stability assessment and safety & speed optimal control that scales or blocks AI actions in the lower layer. • Integration of these building blocks into an HRC+ prototype for collaborative joining tasks (e.g. battery module insertion) as the basis for a “safety-based VLA SDK” for industrial use.			

R&D Target Specifications	The project defines clearly measurable targets to demonstrate the performance of the HRC+ framework not only conceptually but also quantitatively compared to current VLA and RL approaches. The focus is on success rates in planning and execution, reduction of collision-related risks in accordance with ISO/TS 15066, and robust Sim2Real performance on real manipulators and humanoids.
	• **Safe touch & collision risk:**
	  ○ ≥ 90% reduction in risk collision events compared to an unprotected RL baseline controller
	  ○ ≥ 95% compliance with force/energy limits of ISO/TS 15066 in defined test scenarios (with outliers due to sensor noise, not systematic violations)
	• **Prototype performance**: At least one industry-relevant HRC scenario with ≥ 70% overall task success rate under realistic process disturbances and human presence.			

R&D Plan	The R&D plan follows a clearly phase-oriented, hierarchical approach: First, the HRC+ framework is specified and validated in simulation, then transferred to real manipulators and finally ported to a humanoid. Each phase combines methodologically grounded modelling (VLA, RL) with quantitative evaluation (success rates, Sim2Real gap, collision risks), explicitly considering ISO/TS 15066 limits, to deliver a robust, demonstrable human-humanoid collaboration prototype at the end of the project.
	
	Phase 1 (Year 1, Implementation & Simulation)
	• *Germany*: Implementation of VLA task manager and target generator (only in simulation).
	• *Korea*: Development of initial RL motion policies (grasp/move/place/joining) in the simulator.
	• *Korea & Germany*: Establishment of VLA training data pipeline with three complementary sources:
	  • Simulation-based domain randomization (physics, lighting, textures, object variations)
	  • Real-world teleoperation via kinesthetic teaching (hand-guiding on cobot)
	
	Milestone M1 (end of year 1):
	• Architecture functional in simulation; ≥60% task success rate for simple joining tasks in the simulation.			

	Phase 2 (Year 2, real setup & Sim2Real for manipulator) • *Germany*: Implementation and fine-tuning of the VLA task manager (planning, dynamic replanning) and the VLA target generator on a real cobot setup. • *Korea & Germany*: Implementation and validation of the coupling to an initial safety shield. • *Korea*: Development of a Sim2Real pipeline and transfer of RL policies from the simulator to the real manipulator. • *Korea & Germany*: Laboratory demonstrator for collaborative joining tasks with humans in a shared workspace. Milestone M2 (end of year 2): • Reduction of risk collisions by ≥80% compared to RL baseline without shield. **Phase 3 (Year 3, Humanoid & Integrated Prototype)** • *Korea*: Porting the RL policy library and safety shield to a humanoid upper-body in constrained environment. • *Germany*: Integration of dynamic rescheduling logic, VLA target generation and safe touch motion control into an end-to-end system for an industry-relevant scenario. • *Korea & Germany*: Conducting structured benchmark tests (simulation vs. real, humanoid vs. manipulator) with key figure: risk collision reduction ≥90%. • *Korea & Germany*: Derivation of a modular “Safety-based VLA SDK” core (text-to-motion + safety core) as prototype

6. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The German research institute Fraunhofer IGCV is taking on the scientific leadership role in the project for the intelligent collaboration framework (semantic layer) and is ensuring that VLA-based planning and safety logic are implemented and validated in a methodologically grounded and industrially transferable manner. A key strength of the institute is its dual expertise in VLA-based semantic planning and classical motion planning, which enables systematic bridging of the gap between abstract language-based goals and collision-free, executable robot trajectories.
	• **VLA research:** Implementation and LoRA fine-tuning of the VLA task manager and the target generator for low-collision target states; development and evaluation of benchmarks (success rates, Sim2Real gap). Specific methodology:
	  ○ **Data acquisition**: Kinesthetic teaching on UR5/UR5e for 200-500 demonstrations per skill (grasp, move, place, joining), complemented by domain-randomized synthetic data in Isaac Sim
	  ○ **Model validation pipeline**: smolVLA [7] (target generator) and LightPlanner [8] (task planner [9]) for initial data quality assessment and rapid iteration, followed by fine-tuning of OpenVLA [10, 11] or OpenHelix [12] (target generator) for final deployment
	  ○ **Benchmark suite**: Standardized test set of 10 HRC scenarios covering varying lighting, object poses, human proximity levels
	• **Motion planning and interface design**: Support with RL policy training and classical motion planning fallback strategies; Validation of VLA-generated targets via inverse kinematics and collision checking
	• **Experimental validation:** Setting up and operating simulation and laboratory setups (e. g. UR cobot, HRC joining scenarios), systematic data collection for VLA training, and independent evaluation of planning and collaboration performance against defined target metrics.
	• **Resources & dissemination:** Provision of research personnel, simulation and robot infrastructure, data management and scientific documentation (publications, reports) to validate the results and prepare for the subsequent industrialization of the HRC+ framework.

Expected R&R of Korean SME Partner	The Korean SME bears primary responsibility for the development, implementation and demonstration of the lower HRC+ level (Safe Motion Control & Execution) and must therefore demonstrate that RL-based motion planning, Sim2Real transfer and Safety Shield function reliably in real humanoid scenarios under ISO/TS 15066 boundary conditions. **Technical contributions (lower layer):** • Development and implementation of the RL motion policy pool (grasp/move/place/joining, etc.) for cobots and humanoids. • Design and implementation of the safety shield (collision severity/possibility, stability predictor, safety and speed control) including ISO/TS 15066-compliant limit value logic. • Deterministic trajectory scaling/braking based on predicted collision severity and LIDAR safety zone violations All safety-critical functions are implemented as verifiable, non-learning algorithms to enable future certification. **Resources and collaboration:** • Close collaboration with the research institute to implement the interfaces and to jointly evaluate the measurement data. • The Korean SME co-develops the lower HRC+ level (Safe Motion Control & Execution) in close collaboration with Fraunhofer IGCV

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project aims to establish robustly validated foundations for safe human-humanoid collaboration - not just a mere demonstration: The end result will be a two-layer HRC+ framework (VLA-based planning + Safe-RL execution) with quantified safety and performance gains that can be further developed into a marketable "safety-based VLA SDK" for humanoid robots, paving the way for its transition from the laboratory environment to real industrial processes.
	
	Technological results:
	• Integrated HRC+ framework with VLA task manager, target generator, RL policy pool and safety shield, proven to be ISO/TS 15066-compliant in defined HRC scenarios.
	• Safe touch motion control with quantitative reduction of risk collisions compared to RL baselines.
	• Proven Sim2Real chain (Isaac Sim > Cobot > Humanoid) with documented reality gap and robustness limits.
	
	Commercialization potential:
	• Technical basis for a safety-based VLA SDK (text-to-motion + safety core) for retrofitting existing humanoids and cobots.
	• Clear product paths: Licensable middleware, safety engineering toolchain and services for ISO-compliant commissioning.
	• Well-founded data basis for business cases (e. g. reduced teaching time, lower certification costs, higher plant availability).
	
	Industrial and economic impact:
	• Lowering barriers to entry for humanoid robots in high-mix, low-volume production including joining processes; resolving the trade-off between safety and productivity in shared workspaces.
	• Strengthening technological sovereignty in the field of safety intelligence for humanoid systems against US/Chinese solutions.
	
	Scientific added value and limitations:
	• Reference architecture and benchmarks for VLA-based HRC under safety requirements, including documented failure modes and limitations (e.g. sensor quality, model generalisation).
	• Clear research needs after project completion:
	  ○ scaling to more complex processes
	  ○ long-term robustness in 24/7 operation
	  ○ integration of additional sensor technology
	  ○ multi-robot scenarios

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization